

Chapter 2

Wave Functions and Observables

Lecture 5: Expectation Values, Uncertainty, and the Momentum Operator

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Lecture 4 introduced the wave function and the Born rule. This lecture develops the language needed to compare measurement outcomes quantitatively. We first define expectation values and the spread of a probability distribution, then apply these ideas to position and momentum. The main new idea is that momentum is represented in position space by a differential operator.

Questions.

- What does the expectation value of an observable mean when no single measurement gives the average value?
- How should we quantify the spread of possible outcomes?
- Why can position be represented by multiplication by x , while momentum requires a derivative?
- How can the momentum-space probability rule be written entirely in position space?

2.6 Expectation values and uncertainty

2.6.1 Discrete probabilities

Consider a room containing 16 people. Let N_A be the number of people whose age is A :

A	14	15	16	20	21	22
N_A	1	1	3	2	4	5

(2.33)

The probability of selecting a person of age A is

$$P(A) = \frac{N_A}{N_{\text{tot}}}, \quad \sum_A P(A) = 1. \quad (2.34)$$

The most likely age is 22, but the average age is

$$\langle A \rangle = \sum_A A P(A) = \frac{311}{16} \simeq 19.44. \quad (2.35)$$

The average need not be one of the observed values. It is the result that would be approached by averaging the outcomes of many repetitions of the same measurement.

The same rule applies to any function $F(A)$:

$$\langle F(A) \rangle = \sum_A F(A) P(A). \quad (2.36)$$

For example,

$$\langle A^2 \rangle = \sum_A A^2 P(A), \quad \langle A^2 \rangle \neq \langle A \rangle^2 \quad \text{in general.} \quad (2.37)$$

2.6.2 Variance and standard deviation

The deviation of an outcome from the average is $A - \langle A \rangle$. Its average is always zero, so it cannot measure the width of the distribution:

$$\langle A - \langle A \rangle \rangle = 0. \quad (2.38)$$

We therefore average the square of the deviation. This defines the variance and the standard deviation:

$$(\Delta A)^2 \equiv \langle (A - \langle A \rangle)^2 \rangle = \langle A^2 \rangle - \langle A \rangle^2, \quad (2.39)$$

$$\Delta A \equiv \sqrt{\langle A^2 \rangle - \langle A \rangle^2}. \quad (2.40)$$

The expectation value gives the centre of a distribution. The uncertainty ΔA gives its statistical spread. A state with $\Delta A = 0$ gives the same value of A in every measurement.

2.6.3 Continuous probabilities

For a continuous variable X , the probability density $P(x)$ is normalized according to

$$\int_{-\infty}^{\infty} P(x) dx = 1. \quad (2.41)$$

The probability of finding X in a small interval is $P(x) dx$. Consequently,

$$\langle X \rangle = \int_{-\infty}^{\infty} x P(x) dx, \quad (2.42)$$

$$\langle F(X) \rangle = \int_{-\infty}^{\infty} F(x) P(x) dx, \quad (2.43)$$

$$(\Delta X)^2 = \langle X^2 \rangle - \langle X \rangle^2. \quad (2.44)$$

2.7 Position expectation values

For a normalized quantum state $\psi(x)$, the Born rule identifies the position probability density as $P(x) = |\psi(x)|^2$. The expectation value of position is therefore

$$\langle x \rangle_\psi = \int_{-\infty}^{\infty} x |\psi(x)|^2 dx = \int_{-\infty}^{\infty} \psi^*(x) x \psi(x) dx. \quad (2.45)$$

More generally, for any function $f(x)$,

$$\langle f(x) \rangle_\psi = \int_{-\infty}^{\infty} f(x) |\psi(x)|^2 dx. \quad (2.46)$$

In particular,

$$(\Delta x)_\psi^2 = \langle x^2 \rangle_\psi - \langle x \rangle_\psi^2. \quad (2.47)$$

The subscript ψ emphasizes that an expectation value is a property of both the observable and the state. The notation $\langle \psi | x | \psi \rangle$ is another way to write the same expression.

2.7.1 Example: a particle in an interval

Take the normalized state

$$\psi(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{\pi x}{a}\right), \quad 0 < x < a, \quad (2.48)$$

and $\psi(x) = 0$ outside the interval. The probability density is symmetric about $x = a/2$, so

$$\langle x \rangle_\psi = \int_0^a x |\psi(x)|^2 dx = \frac{a}{2}. \quad (2.49)$$

The second moment is

$$\langle x^2 \rangle_\psi = \int_0^a x^2 |\psi(x)|^2 dx = a^2 \left(\frac{1}{3} - \frac{1}{2\pi^2} \right). \quad (2.50)$$

Thus the position uncertainty is

$$\Delta x = a \sqrt{\frac{1}{12} - \frac{1}{2\pi^2}}. \quad (2.51)$$

No single detection has to occur at $x = a/2$. The average position and the uncertainty summarize the outcomes of many position measurements.

2.8 Momentum as an operator

2.8.1 Momentum-space probability

The Fourier expansion from Lecture 4 is

$$\psi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \tilde{\psi}(k) e^{ikx} dk. \quad (2.52)$$

The plane wave e^{ikx} has momentum $p = \hbar k$. Hence $|\tilde{\psi}(k)|^2 dk$ gives the probability of measuring a wave number in $[k, k + dk]$, and

$$\langle p \rangle_\psi = \int_{-\infty}^{\infty} \hbar k |\tilde{\psi}(k)|^2 dk. \quad (2.53)$$

Equivalently, if $\phi(p)$ is normalized in momentum space, then $\langle p \rangle = \int p |\phi(p)|^2 dp$.

There is generally no ordinary function $p(x)$ such that

$$\langle p \rangle = \int_{-\infty}^{\infty} p(x) |\psi(x)|^2 dx. \quad (2.54)$$

Position and momentum are represented in different useful bases. A state of definite position contains many momentum components, while a state of definite momentum is spread over all positions.

2.8.2 The momentum operator

Differentiate a plane wave:

$$\frac{\partial}{\partial x} e^{ikx} = ik e^{ikx}. \quad (2.55)$$

Multiplying by $-i\hbar$ gives

$$-i\hbar \frac{\partial}{\partial x} e^{ikx} = \hbar k e^{ikx} = p e^{ikx}. \quad (2.56)$$

This motivates the position-space momentum operator

$$\hat{p} = -i\hbar \frac{\partial}{\partial x}. \quad (2.57)$$

The operator acts on a wave function. For the Fourier expansion,

$$\hat{p} \psi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hbar k \tilde{\psi}(k) e^{ikx} dk, \quad (2.58)$$

so it multiplies each momentum component by its corresponding momentum $\hbar k$.

2.8.3 The same expectation value in position space

Using the Fourier expansion and the identity

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i(k-k')x} dx = \delta(k - k'), \quad (2.59)$$

we obtain

$$\int_{-\infty}^{\infty} \psi^*(x) \hat{p} \psi(x) dx = \int_{-\infty}^{\infty} \hbar k |\tilde{\psi}(k)|^2 dk = \langle p \rangle_\psi. \quad (2.60)$$

Therefore the momentum expectation value can be calculated without first transforming to momentum space:

$$\langle p \rangle_\psi = \int_{-\infty}^{\infty} \psi^*(x) \hat{p} \psi(x) dx. \quad (2.61)$$

2.9 Operators and measurement statistics

The position and momentum examples illustrate the general rule. Each observable A is represented by an operator $\hat{A}$, and its expectation value in a normalized state is

$$\langle A \rangle_\psi = \int_{-\infty}^{\infty} \psi^*(x) \hat{A} \psi(x) dx. \quad (2.62)$$

For position and momentum,

$$\hat{x} = x, \quad \hat{p} = -i\hbar \frac{\partial}{\partial x}. \quad (2.63)$$

Applying the momentum operator twice gives

$$\hat{p}^2 = -\hbar^2 \frac{\partial^2}{\partial x^2}, \quad \langle p^2 \rangle_\psi = \int_{-\infty}^{\infty} \psi^*(x) \hat{p}^2 \psi(x) dx. \quad (2.64)$$

The uncertainty formula has the same form as before, with moments computed using operators:

$$(\Delta A)^2 = \langle A^2 \rangle - \langle A \rangle^2, \quad \Delta A = \sqrt{\langle A^2 \rangle - \langle A \rangle^2}. \quad (2.65)$$

For physical observables, the operator is Hermitian. This ensures that its expectation value is real. For $\hat{p}$, integration by parts gives the required Hermiticity when the wave function vanishes at the boundaries, or when compatible periodic boundary conditions are imposed.

Questions.

1. Can an expectation value lie between two allowed measurement outcomes? Give a simple example.
2. Why does the average deviation $\langle A - \langle A \rangle \rangle$ always vanish?
3. A wave function is made narrower in position. What happens to its range of momentum components?
4. Verify directly that $\hat{p}e^{ikx} = \hbar k e^{ikx}$.
5. What boundary condition is needed when integrating $\psi^* \hat{p} \psi$ by parts?

Lecture 5 summary

- The expectation value is the probability-weighted average of measurement outcomes.
- The variance and standard deviation quantify the spread: $(\Delta A)^2 = \langle A^2 \rangle - \langle A \rangle^2$.
- For a position-space wave function, $P(x) = |\psi(x)|^2$, so $\langle x \rangle = \int \psi^* x \psi dx$.
- Momentum is read from the Fourier components e^{ikx} , with $p = \hbar k$.
- In position space, momentum is represented by $\hat{p} = -i\hbar \partial_x$.
- The expectation value of any observable is $\langle A \rangle = \int \psi^* \hat{A} \psi dx$.

Suggested reading.

- Griffiths and Schroeter, *Introduction to Quantum Mechanics*, Ch. 1, Secs. 1.3–1.6: *Probability, Normalization, Momentum, and The Uncertainty Principle*. In Sec. 1.3, read both *Discrete Variables* and *Continuous Variables*.
- Feynman, Leighton, and Sands, *The Feynman Lectures on Physics*, Vol. III, Ch. 3, *The Schrödinger Equation in a Classical Context*.
- Review Lecture 4 on Fourier components, plane waves, and the Born rule.